



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Course: CSE391

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Project Title:

PropPilot – Next-Generation Multi-Tenant Property & Asset Management SaaS Platform
(Full-Stack Enterprise Real Estate Operations, Invoicing Engine & Resident Self-Service Portal)

Project Features:

(Core platform functionalities finalized during the project lifecycle)

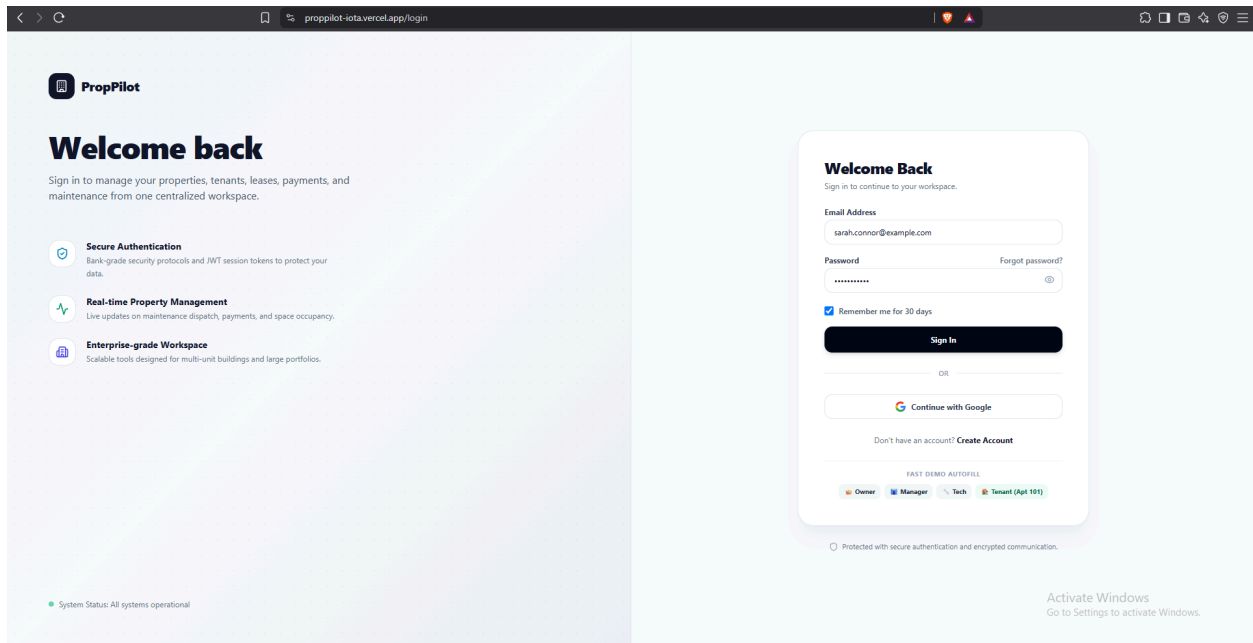
1. **Enterprise-Grade Security & RBAC Framework:**
 - a. Deployment of robust JWT-based authentication utilizing bcryptjs for salted hashing and data encryption.
 - b. Multidimensional Workspace architecture allowing users to navigate and manage several property portfolios concurrently.
 - c. Granular Role-Based Access Control (Owner, Administrator, Manager, Staff, Resident) enforced through secure backend middleware.
 - d. Integrated administrative profile module for identity verification and credential management.
2. **Hierarchical Asset & Spatial Inventory:**
 - a. Structural data modeling of real estate assets: Property Entity → Wing/Floor Segments → Autonomous Rental Units.
 - b. Automated status monitoring for unit lifecycles (Available, Tenanted, Maintenance, or Booked).
 - c. Comprehensive Unit 360° Insight Panel aggregating technical specs, active lease terms, and historical operational data.
3. **Lease Management & Contract Automation:**
 - a. Support for diverse occupancy types including private residential residents and corporate commercial entities.
 - b. End-to-end management of lease states (Draft, Executed, Concluded, Nullified) with configurable deposits and billing logic.
 - c. Cross-functional navigation links connecting financial records directly to physical spatial assets.
4. **Financial Operations & Ledger Analysis:**
 - a. Automated invoicing engine for itemized charges including Rent, Utilities, CAM fees, and Parking.

- b. Real-time receivables tracking categorized by Paid, Pending, Partial, or Default status.
 - c. SSLCommerz gateway integration supporting diverse digital payment methods and automated settlement.
 - d. Strategic financial reporting featuring cash flow visualization, yield analysis, and delinquency monitoring.
- 5. **Maintenance Operations & Work Order Orchestration:**
 - a. Self-service service request interface with domain-specific classification for systems like HVAC and electrical.
 - b. Intelligent dispatch queue tracking tickets through priority levels from Request initiation to Completion.
 - c. Logistics tracking system mapping field service technicians to specific unit-level incidents.
- 6. **Resident Experience & Self-Service Portal:**
 - a. Intuitive resident dashboard providing visibility into financial obligations, contract timelines, and digital documents.
 - b. Seamless unit-switching interface for residents with multiple active rental agreements.
 - c. Frictionless digital payment simulator for instant rent clearing and electronic receipt generation.
- 7. **Digital Repository & Communication Hub:**
 - a. Cloud-based document vault for secure storage of legal agreements, safety certifications, and site plans.
 - b. Broadcasting engine for disseminating critical announcements across specific units or entire portfolios.

Screenshots:

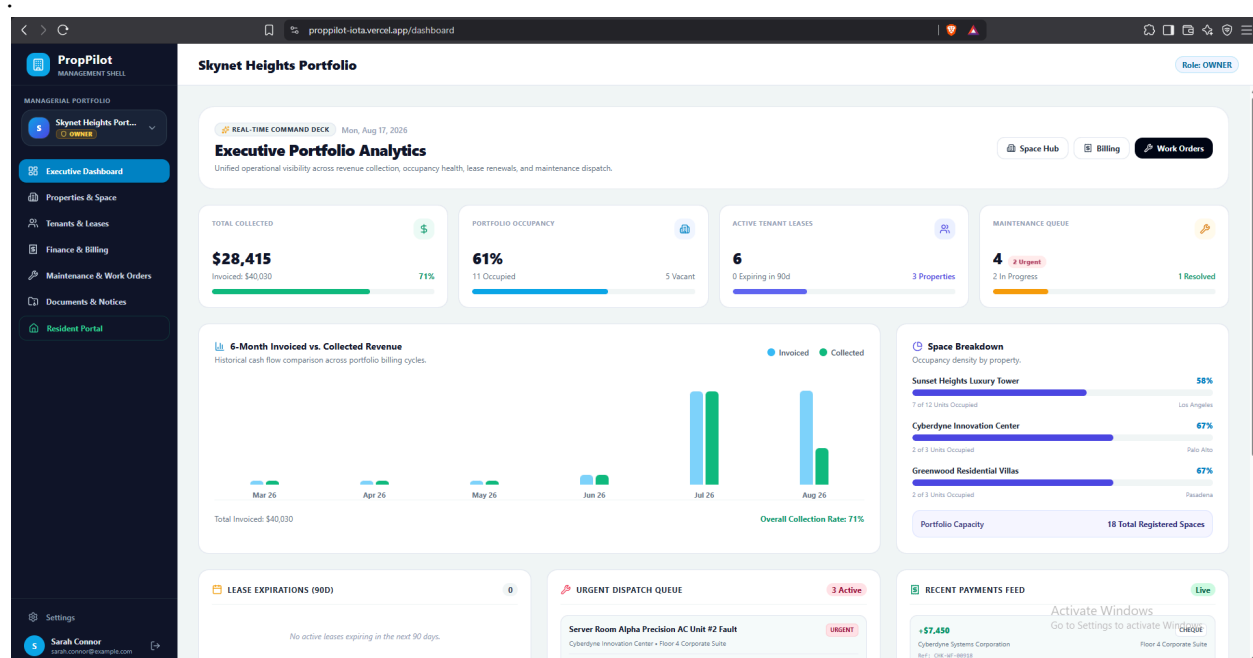
- **Interface Module: Identity Verification & Access Gateway**

Architectural implementation of the login sequence utilizing JWT-based credential validation and automated role-specific routing.



- **Administrative Interface: Centralized Executive Analytics & KPI Oversight**

Architectural visualization of the management console aggregating real-time fiscal performance metrics, portfolio occupancy status, and operational audit trails



- Module Overview: Hierarchical Spatial Inventory & Unit 360° Inspection Panel**
Architectural visualization of real estate assets: Structural Property Entities - Autonomous Units, illustrating occupancy lifecycles and comprehensive technical data aggregation.

Properties & Space Workspace
Manage real estate portfolios, unit groups, rentable space inventory, and 360° operational status.

TOTAL PROPERTIES
3
Active portfolio assets

RENTABLE UNITS
18
Spaces configured

OCCUPANCY HEALTH
61%
11 occupied units

Portfolio Overview (3) | Unit Groups Architecture (7) | **All Rentable Units (18)**

Active Scope: Greenwood Residential Villas | East Garden Villas

Search units...

Unit Name	Property	Unit Group
Villa 101 - Oakwood	Greenwood Residential Villas	East Garden Villas
Villa 102 - Birchwood	Greenwood Residential Villas	East Garden Villas

Villa 101 - Oakwood
UNIT TYPE: 3-Bedroom Royal Villa
CURRENCY / GRACE: USD (5 days)
3-bedroom townhouse with 2-car garage and solar backup.

OUTSTANDING DUE
\$0
\$0 collected • Open Billing...

MAINTENANCE
All Clear
0 total • Open Hub...

ACTIVE OCCUPANT
\$4,200/mo
David Kim
david.kim@example.com
+1-555-0204
Until 12/1/2026
Inspect Tenant Profile in Directory

DIRECT WORKSPACE DEEP-LINES
Leases Hub | Billing Hub | Work Orders | Documents

Activate Windows
Go to Settings to activate Windows.

- Administrative Module: Occupancy Lifecycle & Lease Orchestration Hub**
Architectural visualization of active contractual agreements, integrating resident profiles, fiscal obligations, and cross-functional unit navigation.

Tenants & Leases Workspace
Manage tenant directory profiles, rental lease contracts, space assignments, and occupancy agreements.

TOTAL TENANTS
5
Registered occupant profiles

ACTIVE LEASES
6
Occupied legal contracts

MONTHLY CONTRACTED
\$25,500
Recurring monthly rent

Tenants Directory (5) | **Lease Agreements (6)** | Property Occupancy Hierarchy

Search leases...

TENANT PARTY	ASSIGNED SPACE (PROPERTY - GROUP - UNIT)	MONTHLY RENT
David Kim	Greenwood Residential Villas - East Garden Villas - Villa 101 - Oakwood	\$4,200/mo
Maya Lin	Sunset Heights Luxury Tower - Mid-Rise Panorama (Floors 5-10) - Suite 501	\$2,650/mo
David Kim	Sunset Heights Luxury Tower - Low-Rise Residences (Floors 1-4) - Apt 103	\$2,400/mo
Cyberdyne Systems Corporation	Cyberdyne Innovation Center - Executive Corporate Suites (Floors 4-12) - Floor 4 Corporate Suite	\$12,500/mo
Elena Rostova	Sunset Heights Luxury Tower - Low-Rise Residences (Floors 1-4) - Apt 102	\$1,950/mo
Alex Murphy	Sunset Heights Luxury Tower - Low-Rise Residences (Floors 1-4) - Apt 101	\$1,800/mo

ASSIGNED TENANT
David Kim
david.kim@example.com

ASSIGNED SPACE HIERARCHY
Greenwood Residential Villas - East Garden Villas - Villa 101 - Oakwood
Unit ID: 3K64C03...
Inspect Apartment

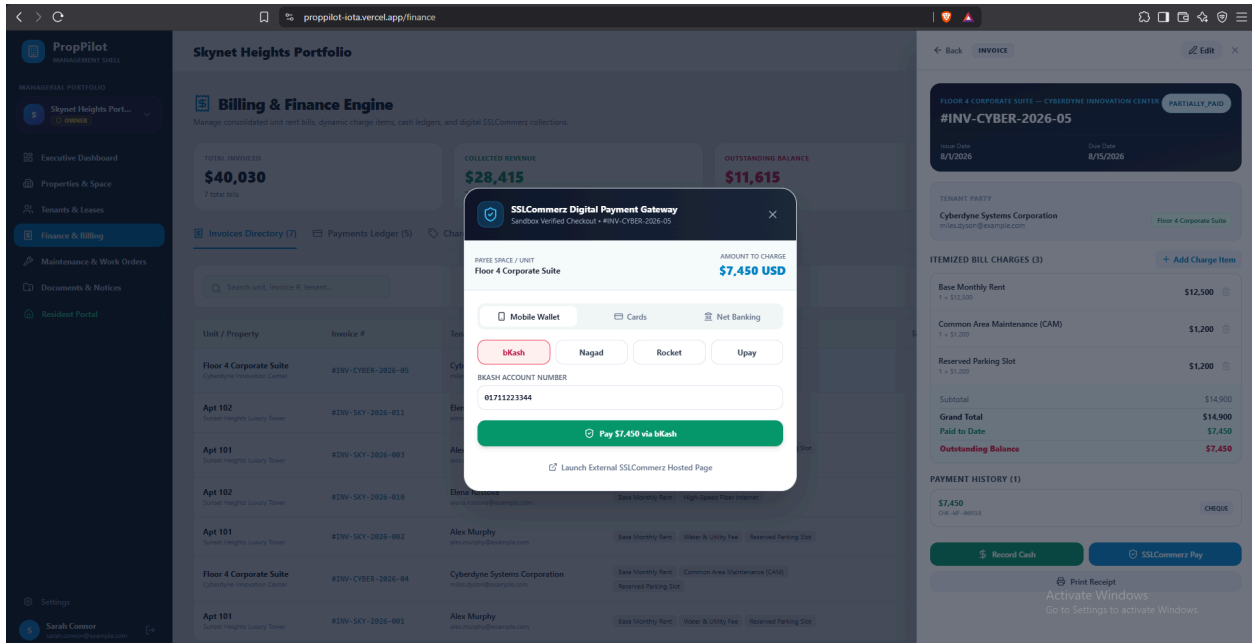
MONTHLY RENT
\$4,200

SECURITY DEPOSIT
\$8,400

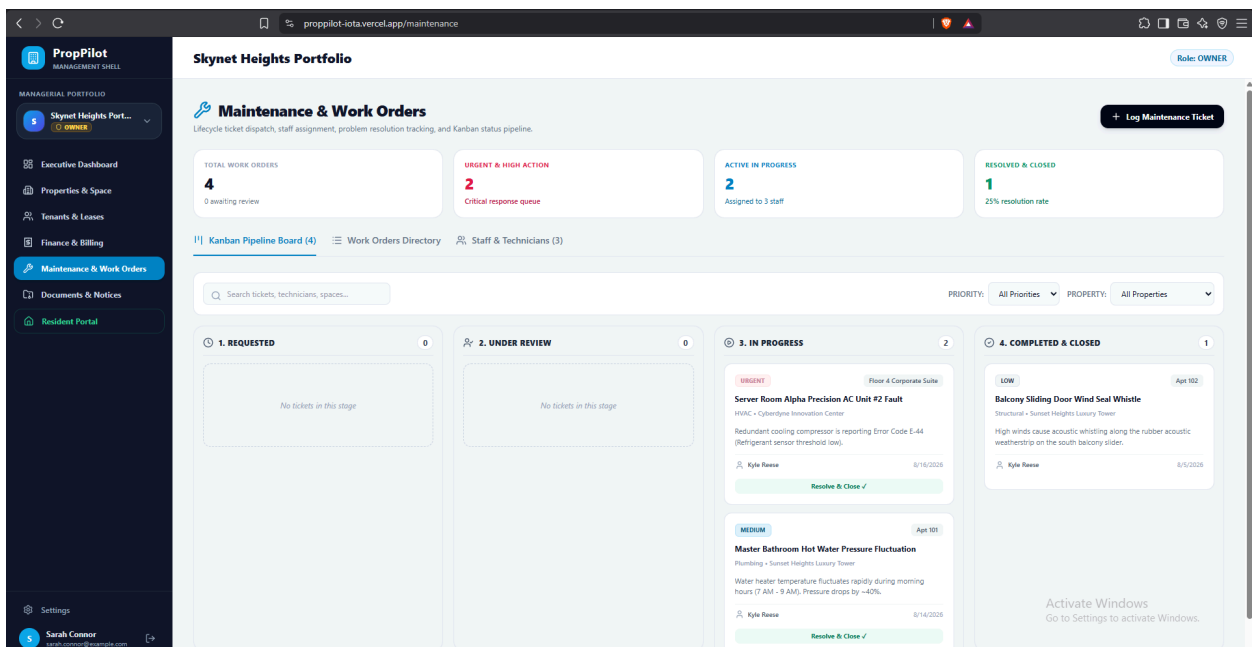
CONTRACT DURATION
12/1/2025 - 12/1/2026
Inspect Apartment in Properties Page

Activate Windows
Go to Settings to activate Windows.

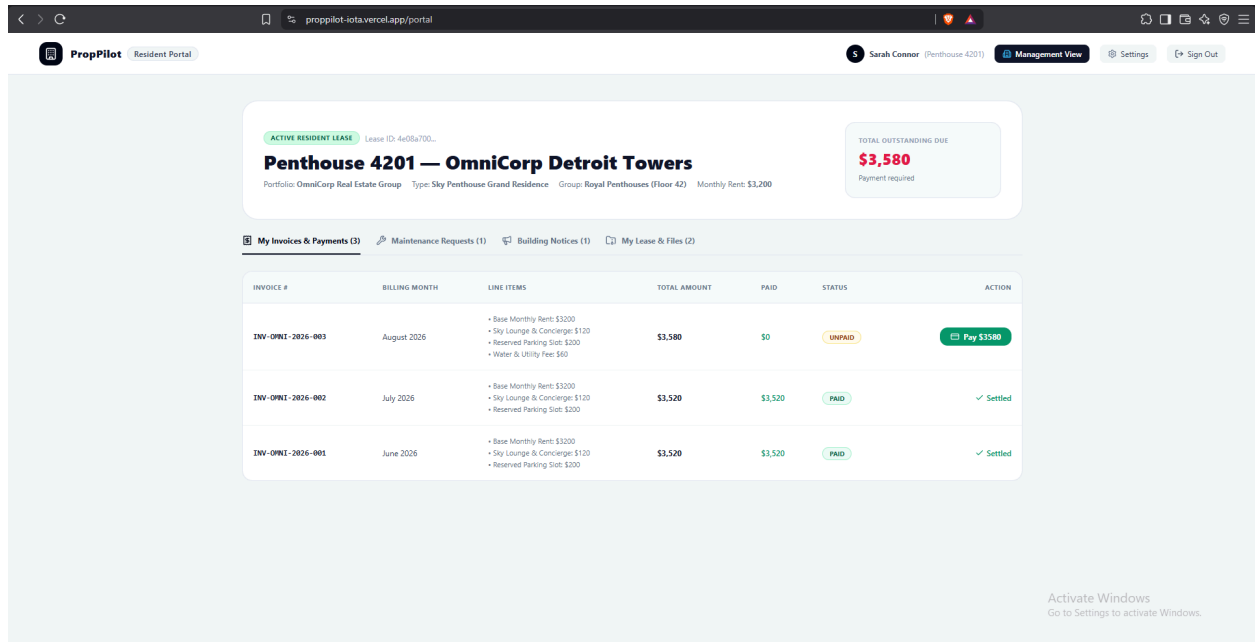
- **Financial Interface: Itemized Ledgering & SSLCommerz Gateway Integration**
Systematic presentation of the automated invoicing engine alongside the digital payment settlement environment and sandbox checkout sequence.



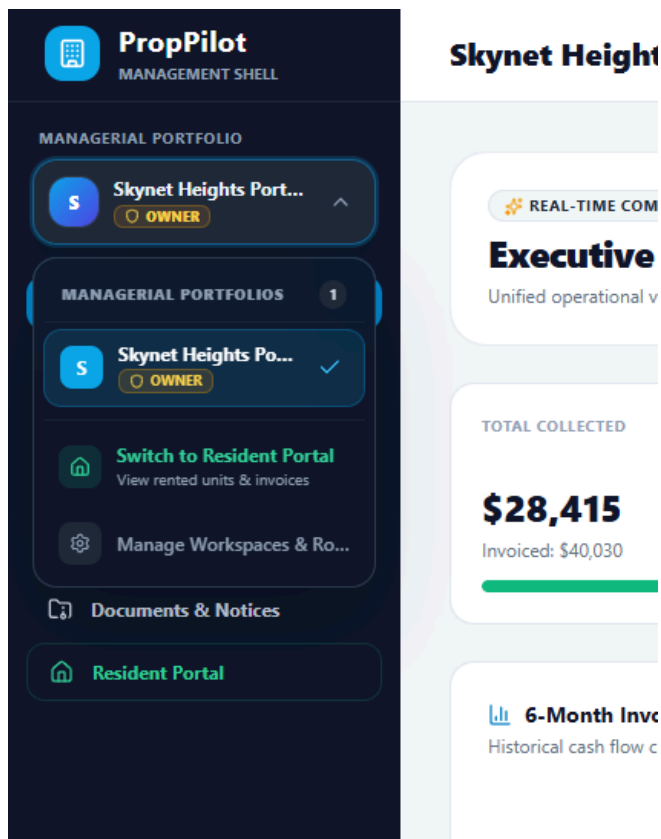
- **Operational Dashboard: Maintenance Orchestration & Service Dispatch Tracking**
Intelligent ticketing board featuring multi-tier priority classification, domain-specific filtering, and real-time logistics for technical field services.



- Resident Portal: Tenant Self-Service Experience & Digital Interaction Gateway (/portal)**
End-user interface enabling seamless unit-switching for multi-lease residents, frictionless rent clearing, and automated service request initiation.



- Interface Component: Multidimensional Glassmorphism Portfolio Navigation**
Architectural implementation of the workspace switcher utilizing a modern glassmorphism aesthetic for concurrent management of several property portfolios.



Online Resources used:

Official Documentation & Learning References:

- a) **MDN Web Docs:** Exploration of HTML5 semantic structure, JavaScript module systems, and asynchronous programming paradigms (<https://developer.mozilla.org>)
- b) **React 19 Documentation:** Mastering component lifecycles, essential React Hooks (useState, useEffect, useRef, useContext), and the Virtual DOM (<https://react.dev>)
- c) **Prisma ORM:** Implementation of relational data modeling, automated schema migrations, and type-safe client queries (<https://www.prisma.io/docs>)
- d) **Vite.js:** Advanced frontend build tooling, production optimization, and development performance (<https://vitejs.dev/guide>)
- e) **Tailwind CSS:** Designing responsive interfaces using utility-first CSS and dynamic variables (<https://tailwindcss.com/docs>)
- f) **PostgreSQL:** Architecting relational schemas, ensuring referential integrity, and database indexing (<https://www.postgresql.org/docs>)
- g) **Express.js:** Building RESTful APIs with complex routing, middleware, and error management (<https://expressjs.com>)
- h) **W3Schools:** Reference for web standards, Flexbox/Grid layouts, and standard HTTP protocols (<https://www.w3schools.com>)

Developer Community & Open Source Repositories:

- a) **Stack Overflow:** Resolving CORS issues, database deployment hurdles, and routing state management logic.
- b) **Lucide Icons:** Integration of an open-source visual asset toolkit (<https://lucide.dev>)
- c) **GitHub Repository:** Primary codebase for the PropPilot project (<https://github.com/izazalamz/proppilot>)

Future Enhancements:

(Strategic roadmap for enhancing platform scalability, intelligence, and overall user interaction)

- 1) **AI-Driven Lease & Document Extraction:** Leveraging OCR and Large Language Models for automated parsing of PDF agreements to facilitate seamless data population.
- 2) **Integrated Messaging & Notification Hub:** Utilizing Twilio and WhatsApp APIs for automated dispatch of payment reminders, receipts, and critical maintenance alerts.
- 3) **IoT-Enabled Infrastructure Management:** Connecting digital locks and smart metering systems to enable remote property access and automated utility billing workflows.
- 4) **Native Cross-Platform Mobility:** Engineering React Native applications for iOS and Android to support push notifications and mobile-first maintenance reporting.
- 5) **Global Transaction & Payment Gateway:** Incorporating Stripe, PayPal, and ACH transfers with dynamic multi-currency conversion for international financial operations.
- 6) **Predictive Intelligence & Data Forecasting:** Deploying machine learning to analyze tenant churn, optimize rental yields, and establish proactive maintenance schedules.